

Zenith AI Sandbox - Test Knowledge Base

Official Evaluation & Verification Document for RAG Testing

1. Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is an architectural pattern that connects Large Language Models (LLMs) to external domain-specific knowledge bases. By retrieving relevant document passages at query time, RAG eliminates LLM hallucinations and provides up-to-date factual answers without expensive fine-tuning.

2. Chroma Vector Database & Storage Engine

Chroma DB is an open-source, AI-native vector database. It converts text chunks into 384-dimensional mathematical vector embeddings using MD5 feature hashing algorithms. Under the hood, Chroma stores document strings, unique UUID row IDs, and dense vector arrays in a local SQLite file named `chroma.sqlite3`.

3. Zenith System Specifications & Parameters

The Zenith AI System operates with a default chunk size of 200 characters and a chunk overlap of 40 characters. It utilizes cosine similarity matching for dense semantic vector search and streams real-time generative responses using the Gemini 3.6 Flash model.